STAGE 1: Decentralized Federated Self-Supervised Pre-Training (FedProx-MAE on ViT-Tiny)

Privacy-Preserving Pre-Training on Uncurated Chest Radiographs (No Raw Patient Data Shared Across Hospital Silos)

Hospital Clients (N = 10)

Non-IID Unlabeled Silos

- Client 1: NIH Silo (n=12,286)
- Client 2: NIH Silo (n=7,738)
- Client 3: NIH Silo (n=4,136)
- Client 10: NIH Silo (n=308)
  (Dirichlet Non-IID Skew: α=0.5)

Local Client: Masked Autoencoder (MAE)

ViT-Tiny Architecture (5.7M Parameters)

Input Radiograph

x in R^(224 x 224 x 3) (Chest X-Ray)

Patch Extraction & Masking

Patch: 16x16 (N=196), Masking Ratio: 75%
Visible: 49 patches, Masked: 147 patches

ViT-Tiny Encoder (f_theta)

12 Transformer Blocks, Embed Dim=192, 3 Heads
Processes only unmasked visible patches

Lightweight Decoder & Loss

4 Blocks, Dim=128 | MSE on Masked Patches
L_FedProx = L_MAE + (mu/2) ||w_k - w^t||^2 (mu=0.01)

Local w_k

Global w^(t)

Federated Central Server

FedAvg Coordination (R = 100 Rounds)

Parameter Aggregation

w^(t+1) = sum_(k=1)^10 (n_k / n) w_k^(t+1)
Sample-weighted parameter averaging

Zero Data Transfer

Raw X-rays never leave local nodes.
Only encoder/decoder tensors synced.

Global Model Broadcast

Broadcast global w^(t+1) to clients.
Converges from loss 0.658 -> 0.133.

Transfer Pretrained Global Encoder Weights theta* to Edge Clinics

STAGE 2: Low-Resource Few-Shot Metric Adaptation & Clinical Triage (Prototypical Network)

Zero-Backpropagation Edge Clinic Deployment with Extreme Label Scarcity (K = 5 to 20 Reference Shots)

Scarce Clinical Data (K-Shot)

Support Episodes (K=5 to 20 per class)

Normal Reference (K=5-20)

Confirmed healthy chest X-rays

TB Reference (K=5-20)

Microbiologically confirmed TB

Frozen Metric Space & Prototypes

Zero Backpropagation / Linear Projection

Frozen Encoder f_theta*

Extracts 192-dim latent vector: z = f_theta*(x)

Prototypical Class Centroids

c_Normal = (1/K) sum z_i | c_TB = (1/K) sum z_j

Query Classification & Evaluation

Cosine Metric Distance Softmax

Shenzhen In-Domain Query

AUC: 81.26% | Acc: 77.49% | Sens: 78.48%
(Outperforms ImageNet baseline by +4.5%)

Montgomery Held-Out Stress Test

Cross-Hospital Domain Shift Quantified
(Preserves high diagnostic sensitivity)